

Network Traffic Visualization & Statistics Report

Milestone 3: Detection & Visualization

1. Introduction

Milestone 3 focused on converting the network traffic observations obtained during the previous PCAP analysis into structured detection logic, statistical analysis, visual representations, and an overall security investigation assessment.

During Milestone 2, the network traffic investigation was performed across three separate PCAP-derived CSV traffic segments, each representing a portion of the overall captured network traffic.

Before beginning the combined statistical analysis and visualization process, the three Milestone 2 datasets were consolidated into a single dataset using a Python script. This ensured that the analysis performed during Milestone 3 was based on the complete available traffic dataset rather than treating each segment independently.

The three traffic segments contained the following record counts:

Segment	Records
Traffic Segment 1	1,200
Traffic Segment 2	1,200
Traffic Segment 3	1,188
Combined dataset	3,588

The resulting combined file was named Combined_Traffic_Data.csv. This combined dataset was then used as the common source for statistical analysis, detection-rule development, traffic visualization, anomaly identification, comparative analysis, and the final analyst assessment.

The overall workflow followed was:

```
-> Milestone 2 PCAP Segments
-> Python Verification
-> Python Combination Script
-> Combined Dataset
-> Statistical Analysis
-> Detection Rules
-> Visualizations
-> AI-Assisted Interpretation
-> Analyst Verification
-> Comparative Analysis
-> Final Security Assessment
```

2. Objectives of Milestone 3

The objectives of this milestone were to:

1. Identify consistent indicators of potential security concern from the previous traffic analysis.
2. Develop three to five detection rules using IF/THEN style detection logic.
3. Explain why each detection rule is relevant and identify possible false positives.
4. Create network traffic visualizations covering traffic flow, protocol distribution, destination port usage, top communicating hosts, external communication, routine versus investigation-relevant traffic, frame-size behaviour, detection-rule mapping, and overall investigation status.
5. Highlight anomalies and investigation-relevant traffic patterns.
6. Compare routine/observed traffic with investigation-relevant behaviour.
7. Use AI to assist in analysing and interpreting the visual data.
8. Independently verify the AI-generated reasoning against the established statistical findings.
9. Compile the visuals, statistical findings, detection rules, and interpretations into the final report.

3. Source Data Preparation and Combination

3.1 Milestone 2 Traffic Segments

The investigation began with three PCAP-derived CSV datasets produced during Milestone 2. The datasets were kept as separate segments during the initial network traffic analysis; for Milestone 3, a combined dataset was required so that statistical calculations and visualizations represented the complete traffic investigation.

The three segment counts reconcile to the final dataset: $1,200 + 1,200 + 1,188 = 3,588$ records.

3.2 Python-Based Dataset Combination

Python was used to automate the dataset combination rather than manually copying and pasting the traffic records. The Python process used the three traffic segment CSV files as input, added source-file information to each record, and combined the records into a single dataset.

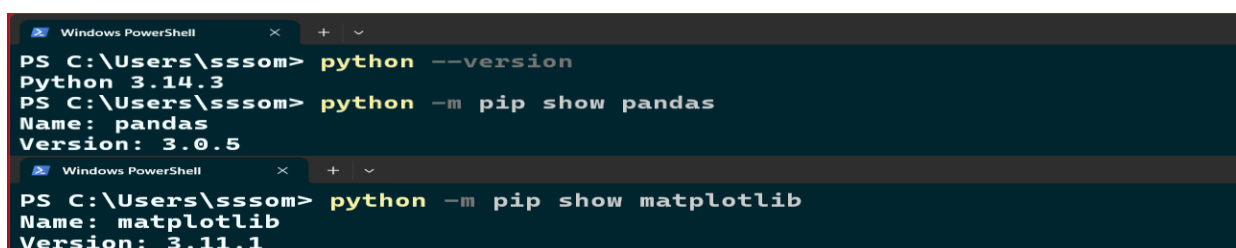
Combination script: combine_traffic_data.py

Output file: Combined_Traffic_Data.csv

The successful result contained 3,588 records, providing one consistent dataset for all subsequent Milestone 3 calculations and visualizations.

3.3 Python Verification Evidence

Two screenshots were retained as evidence of this preparation process.

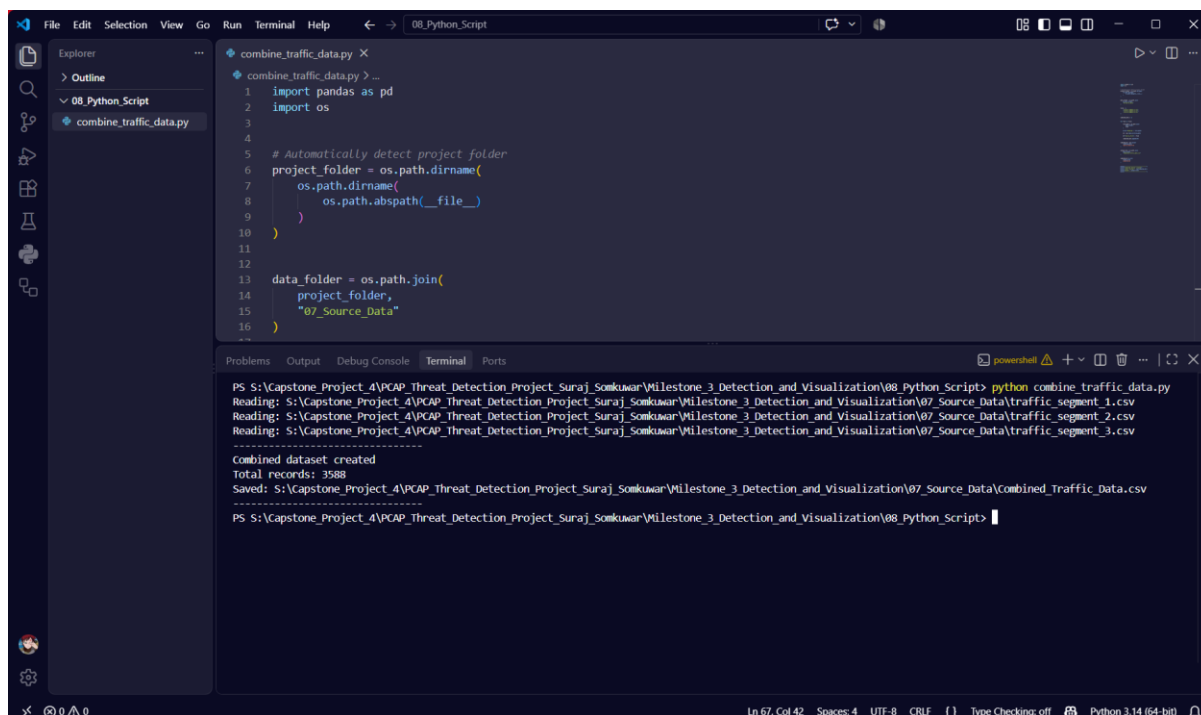


```
Windows PowerShell
PS C:\Users\ssom> python --version
Python 3.14.3
PS C:\Users\ssom> python -m pip show pandas
Name: pandas
Version: 3.0.5

Windows PowerShell
PS C:\Users\ssom> python -m pip show matplotlib
Name: matplotlib
Version: 3.11.1
```

Figure: Python environment/library verification

This screenshot documents the Python preparation and verification stage performed before the traffic datasets were combined.



The screenshot shows a Visual Studio Code editor with a Python script named `combine_traffic_data.py` open. The script uses the `os` and `pandas` modules to automatically detect the project folder and combine three CSV files into a single dataset. The terminal window below the editor shows the command `python combine_traffic_data.py` being executed, followed by the output showing the paths of the three source CSV files, the creation of the combined dataset, the total number of records (3588), and the path where the combined dataset was saved.

```
combine_traffic_data.py
1 import pandas as pd
2 import os
3
4
5 # Automatically detect project folder
6 project_folder = os.path.dirname(
7     os.path.dirname(
8         os.path.abspath(__file__)
9     )
10 )
11
12
13 data_folder = os.path.join(
14     project_folder,
15     "07_Source_Data"
16 )
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```

```
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\08_Python_Script> python combine_traffic_data.py
Reading: S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\07_Source_Data\traffic_segment_1.csv
Reading: S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\07_Source_Data\traffic_segment_2.csv
Reading: S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\07_Source_Data\traffic_segment_3.csv
-----
Combined dataset created
Total records: 3588
Saved: S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\07_Source_Data\Combined_Traffic_Data.csv
-----
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone_3_Detection_and_Visualization\08_Python_Script>
```

Figure: Python script used to combine the three traffic segments

This screenshot provides evidence that the three Milestone 2 traffic segments were programmatically combined before the statistical and visualization analysis began.

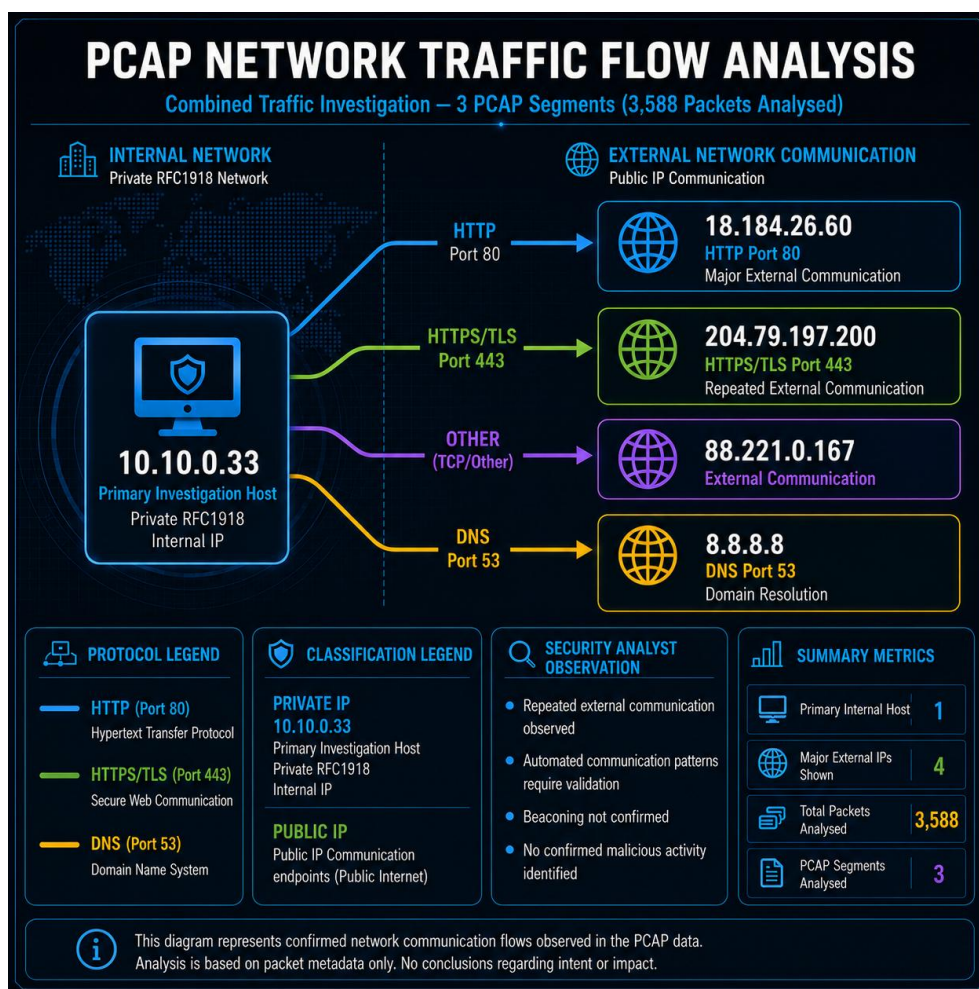
3.4 Combined Dataset Validation

After combination, the resulting dataset contained 3,588 total records, made up of 3,549 IP-bearing records and 39 ARP or non-IP records. The reconciliation is straightforward: $3,549 + 39 = 3,588$ total records.

Timestamps were available for all 3,588 records. The captured traffic period represented in the combined dataset ran from 2021-08-17 01:58:06.017228 +0530 to 2021-08-17 02:08:16.417023 +0530, a duration of approximately 10 minutes and 10.40 seconds.

The primary internal/private host identified throughout the investigation was 10.10.0.33. The dominant external/public-unicast endpoint was 18.184.26.60.

4. Network Traffic Flow Analysis



Visual 1: PCAP Network Traffic Flow Analysis

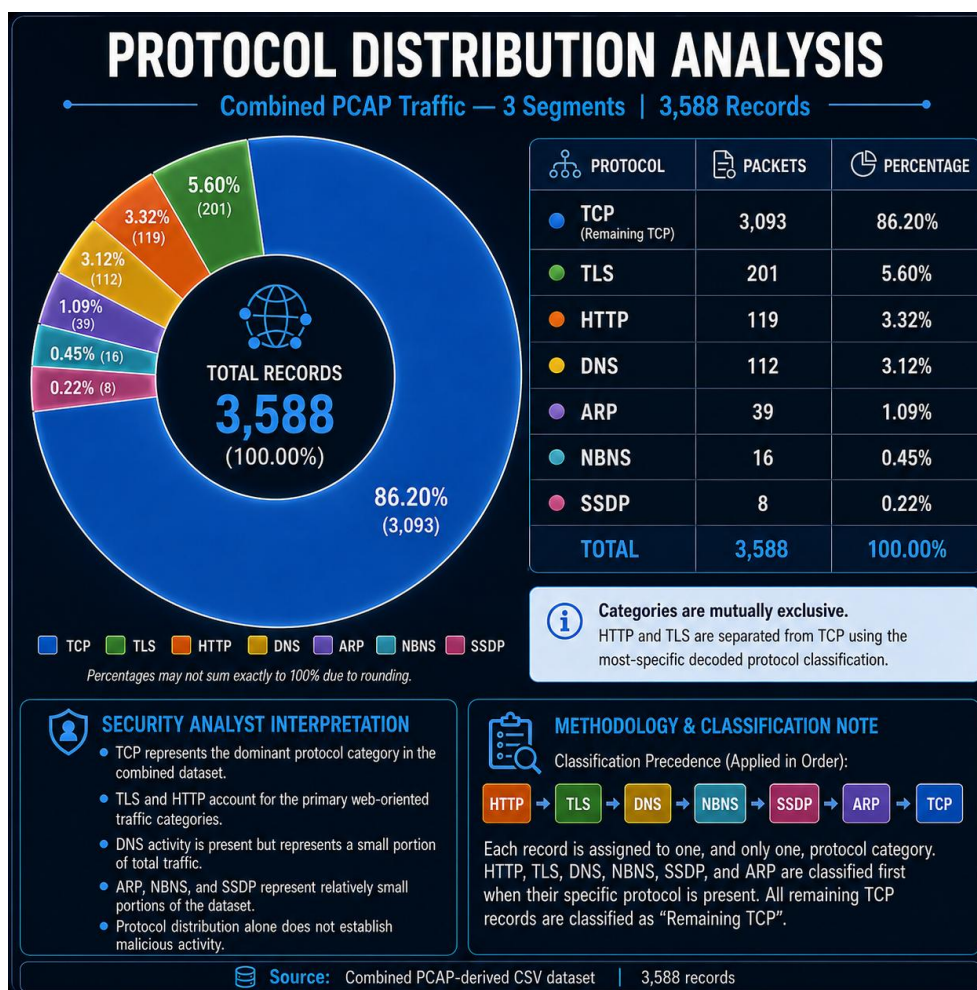
Interpretation

The network traffic flow visualization provides a high-level representation of the communication relationships observed in the combined PCAP-derived dataset. The primary internal/private host was 10.10.0.33, an address within the private RFC1918 address space and the principal internal host involved in the observed public-unicast communication.

The visualization identifies several external/public communication endpoints, including 18.184.26.60, 204.79.197.200, 88.221.0.167, and 8.8.8.8. It also identifies the principal communication categories: HTTP (TCP/80), HTTPS/TLS (TCP/443), other TCP communication, and DNS (UDP/53).

The diagram demonstrates that communication between the internal host and external/public endpoints was a significant characteristic of the captured traffic. However, the traffic-flow visualization represents observed packet metadata and communication relationships only; it does not, by itself, establish malicious intent, compromise, command-and-control activity, or data exfiltration.

5. Protocol Distribution Analysis



Visual 2: Protocol Distribution Analysis

5.1 Protocol Statistics

The combined dataset contained the following mutually exclusive protocol categories:

Protocol	Records	Percentage
TCP (remaining)	3,093	86.20%
TLS	201	5.60%
HTTP	119	3.32%
DNS	112	3.12%
ARP	39	1.09%
NBNS	16	0.45%
SSDP	8	0.22%
Total	3,588	100.00%

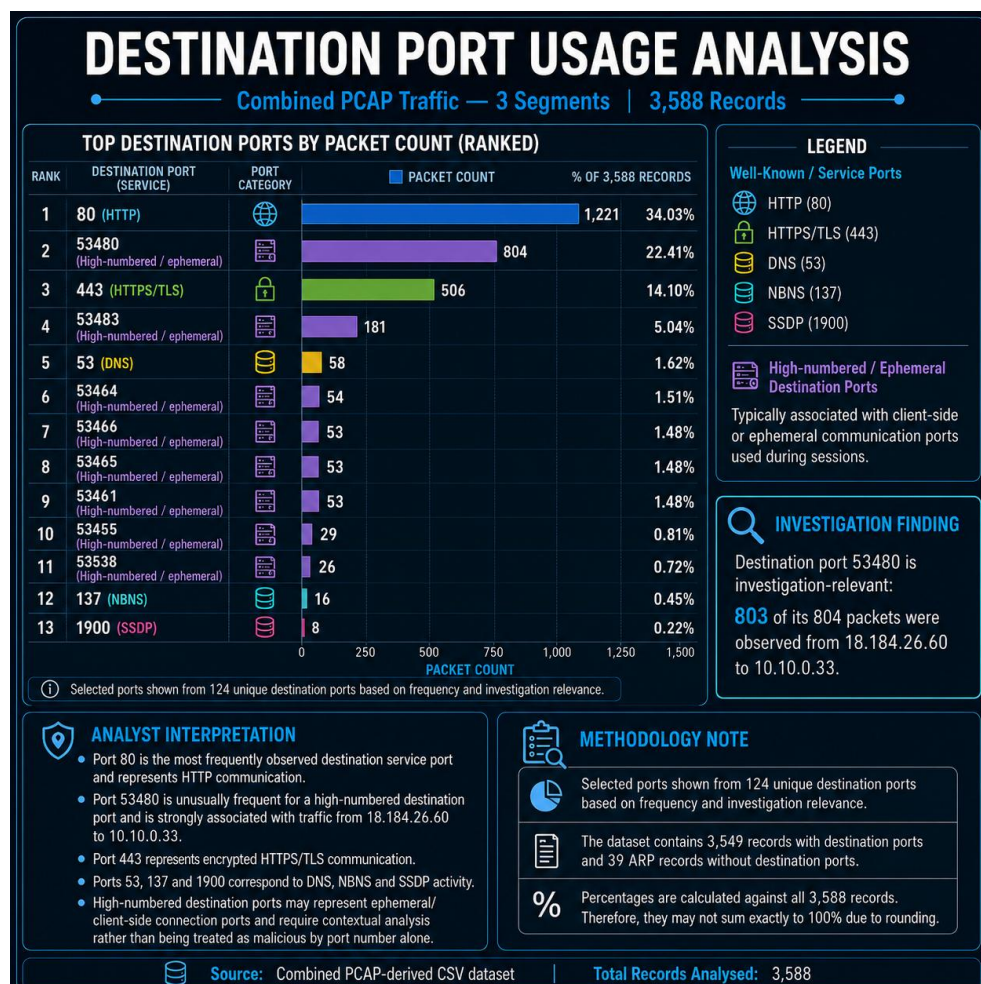
5.2 Methodology and Classification Note

The protocol categories were treated as mutually exclusive. The classification precedence applied in the analysis was HTTP, then TLS, then DNS, then NBNS, then SSDP, then ARP, then remaining TCP: when a record contained a specific decoded protocol, that protocol was classified first, and all remaining TCP records were classified as TCP (remaining). This approach prevents the same record from being counted in more than one protocol category.

5.3 Interpretation

TCP was the dominant protocol category, accounting for 3,093 records (86.20%). TLS represented 201 records (5.60%), HTTP represented 119 records (3.32%), and DNS represented 112 records (3.12%). ARP, NBNS, and SSDP represented smaller portions of the dataset. The protocol distribution demonstrates the overall composition of the captured traffic, but protocol type alone does not establish malicious activity.

6. Destination Port Usage Analysis



Visual 3: Destination Port Usage Analysis

The destination-port analysis examined the frequency of observed destination ports. The major destination-port observations were:

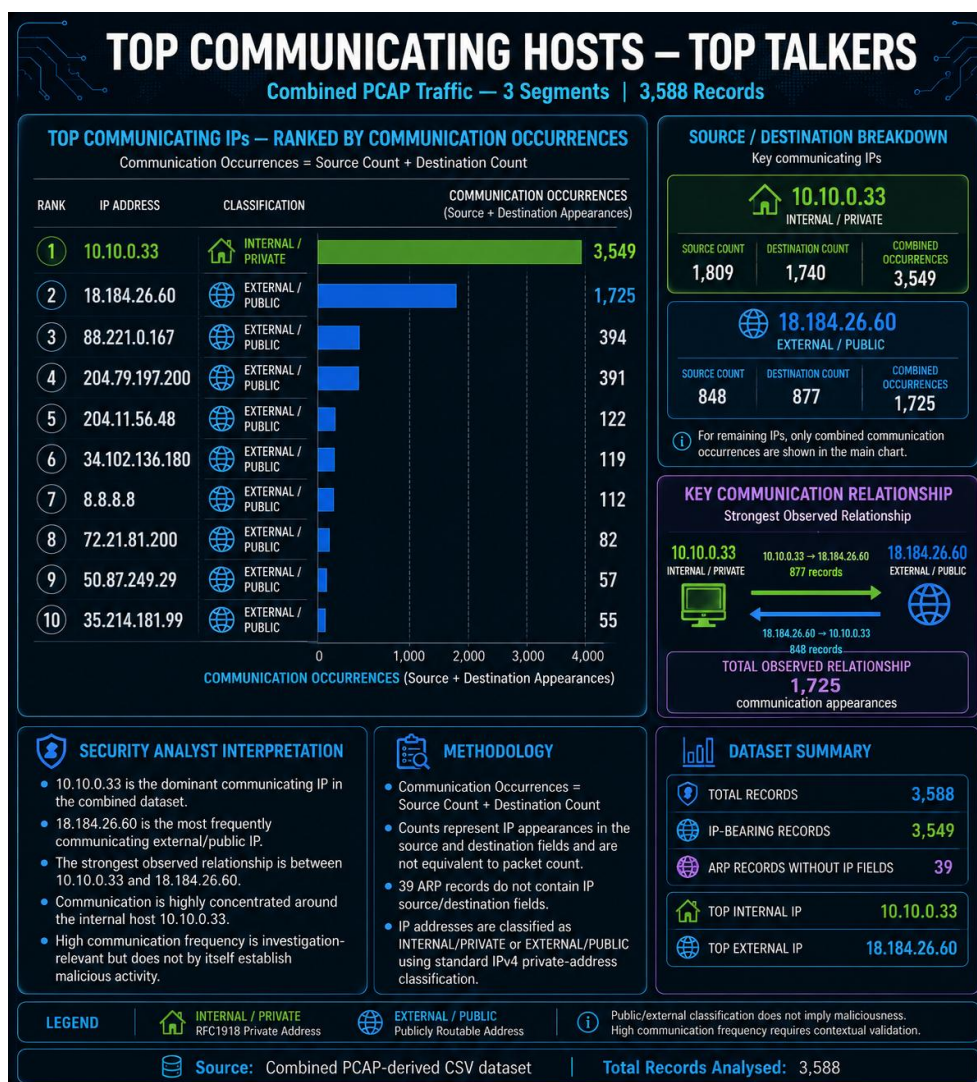
Destination Port	Records	% of All Records
TCP/80	1,221	34.03%
TCP/53480	804	22.41%
TCP/443	506	14.10%
TCP/53483	181	5.04%
UDP/53	58	1.62%
UDP/53464	54	1.51%
UDP/53466	53	1.48%
UDP/53465	53	1.48%
UDP/53461	53	1.48%
UDP/53455	29	0.81%
UDP/53538	26	0.72%
UDP/137	16	0.45%
UDP/1900	8	0.22%

The visual states that the dataset contained 124 unique destination ports, with the ports above displayed according to frequency and investigation relevance.

Important TCP/53480 Finding

One of the strongest destination-port observations was TCP/53480. The dataset contained 804 TCP/53480 records, representing 22.41% of all records. All 804 of these records were observed as 18.184.26.60:80 to 10.10.0.33:53480, with zero exceptions. This relationship was therefore considered investigation-relevant because of its frequency and concentration. However, TCP/53480 is a high-numbered, client-side destination port, and the port number alone does not establish malicious activity.

7. Top Communicating Hosts (Top Talkers)



Visual 4: Top Communicating Hosts

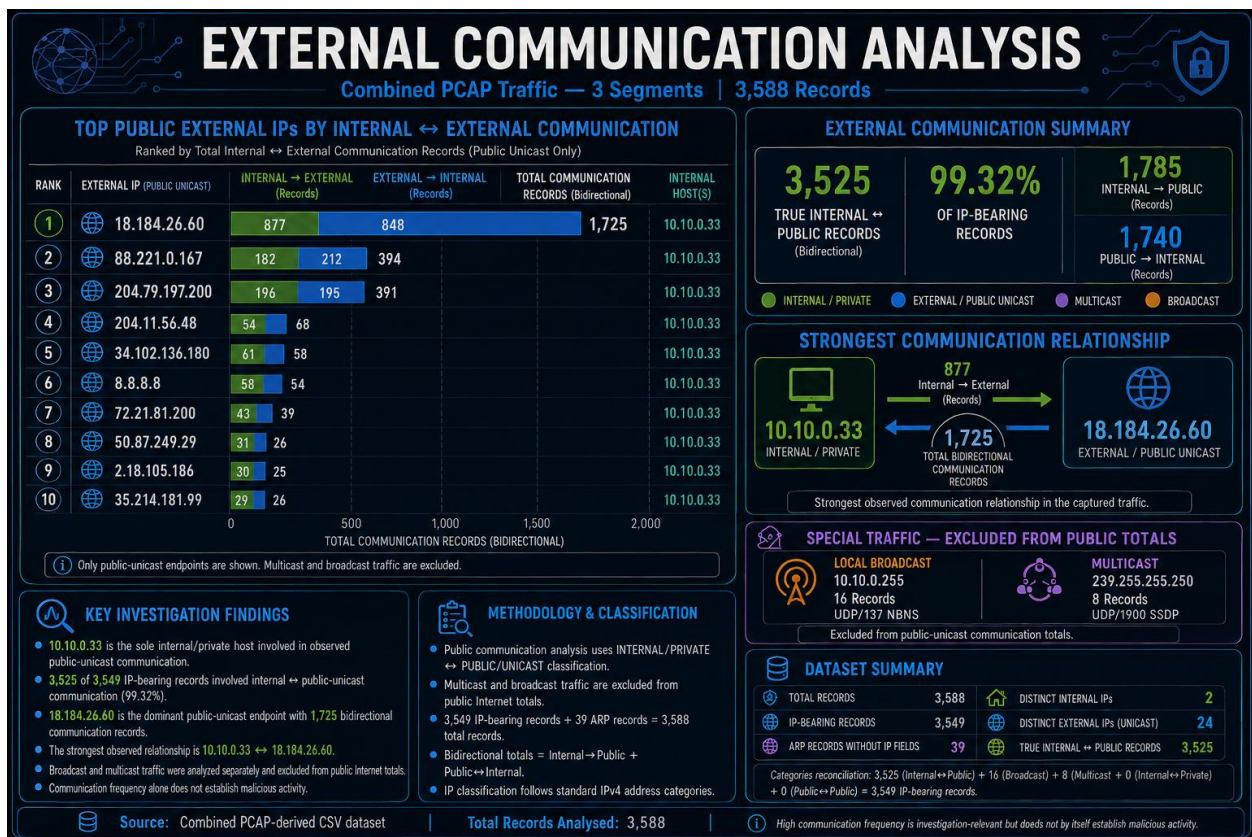
The visualization defines communication occurrences as the sum of source count and destination count. Using this metric, the highest communication occurrences were:

Rank	IP Address	Classification	Occurrences
1	10.10.0.33	Internal / Private	3,549
2	18.184.26.60	External / Public	1,725
3	88.221.0.167	External / Public	394
4	204.79.197.200	External / Public	391
5	204.11.56.48	External / Public	122
6	34.102.136.180	External / Public	119
7	8.8.8.8	External / Public	112
8	72.21.81.200	External / Public	82

Rank	IP Address	Classification	Occurrences
9	50.87.249.29	External / Public	57
10	35.214.181.99	External / Public	55

The primary internal host, 10.10.0.33, was therefore the dominant communicating IP in the combined dataset. The strongest external relationship was 10.10.0.33 to 18.184.26.60, with 1,725 communication records. This high communication frequency was treated as investigation-relevant, but not as proof of malicious activity.

8. External Communication Analysis



Visual 5: External Communication Analysis

External Communication Finding

The public-unicast traffic analysis produced the following key statistics: 1,785 internal-to-public records and 1,740 public-to-internal records, for a combined total of 3,525 records. This represents 98.24% of all records and 99.32% of IP-bearing records. The calculation is straightforward: 1,785 + 1,740 = 3,525, and these 3,525 records represent true internal-to-public-unicast communication.

The dominant communication relationship was 10.10.0.33 to 18.184.26.60, with 1,725 records. The directional breakdown of this relationship was 877 records from 10.10.0.33 to 18.184.26.60 and 848 records from 18.184.26.60 to 10.10.0.33, which together reconcile to 877 + 848 = 1,725.

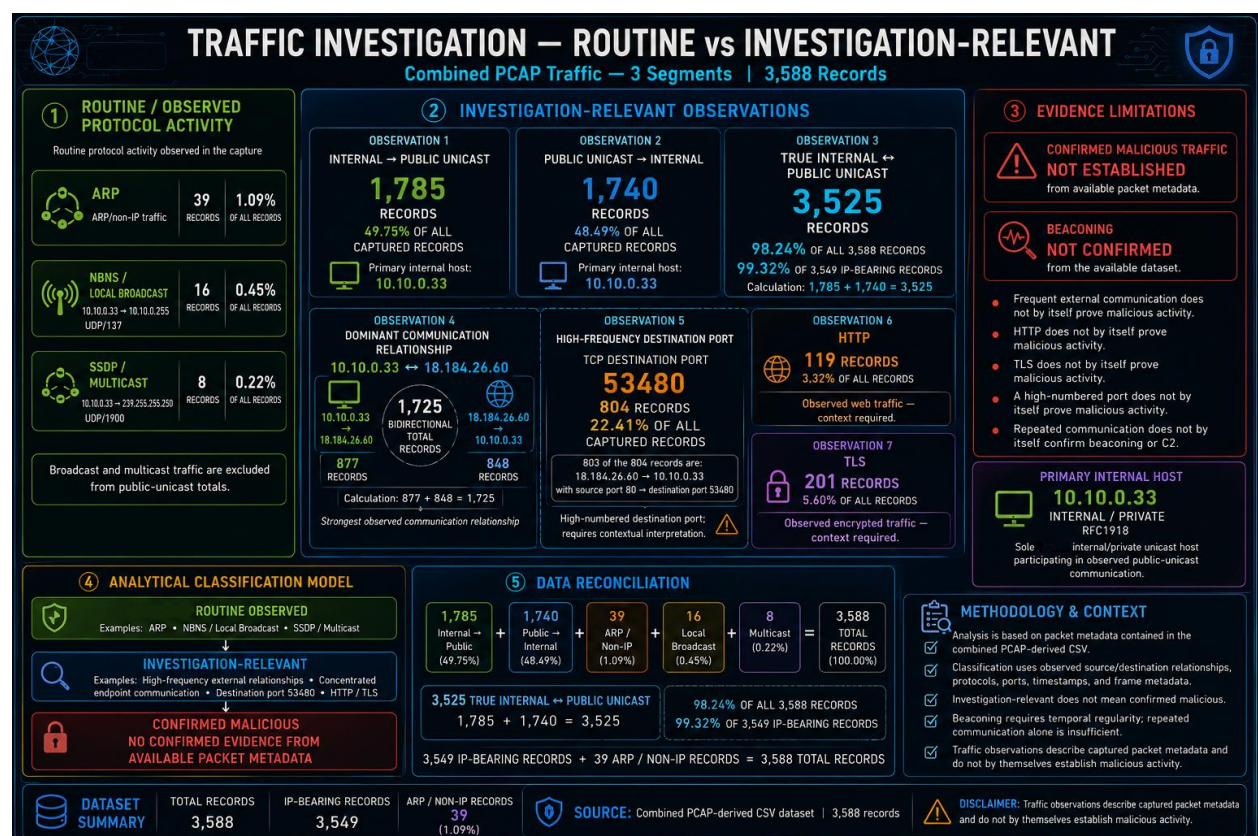
Special Traffic Excluded from Public-Unicast Totals

The analysis deliberately separated broadcast and multicast traffic from public-unicast communication. Local broadcast traffic from 10.10.0.33 to 10.10.0.255, using NBNS over UDP/137, accounted for 16 records and was classified as routine, observed special traffic. Multicast traffic from 10.10.0.33 to 239.255.255.250, using SSDP over UDP/1900, accounted for 8 records and was classified the same way. Both categories were excluded from the public-unicast communication totals.

Interpretation

The external communication analysis demonstrates that the majority of IP-bearing traffic involved internal-to-public-unicast communication. However, communication frequency alone does not establish malicious activity. The external endpoints should therefore be validated using additional context such as asset ownership, expected application behaviour, threat intelligence, and historical network baseline.

9. Routine vs Investigation-Relevant Traffic



Visual 6: Traffic Investigation: Routine vs Investigation-Relevant

The comparative analysis separated observed traffic into routine/special-purpose activity and investigation-relevant observations.

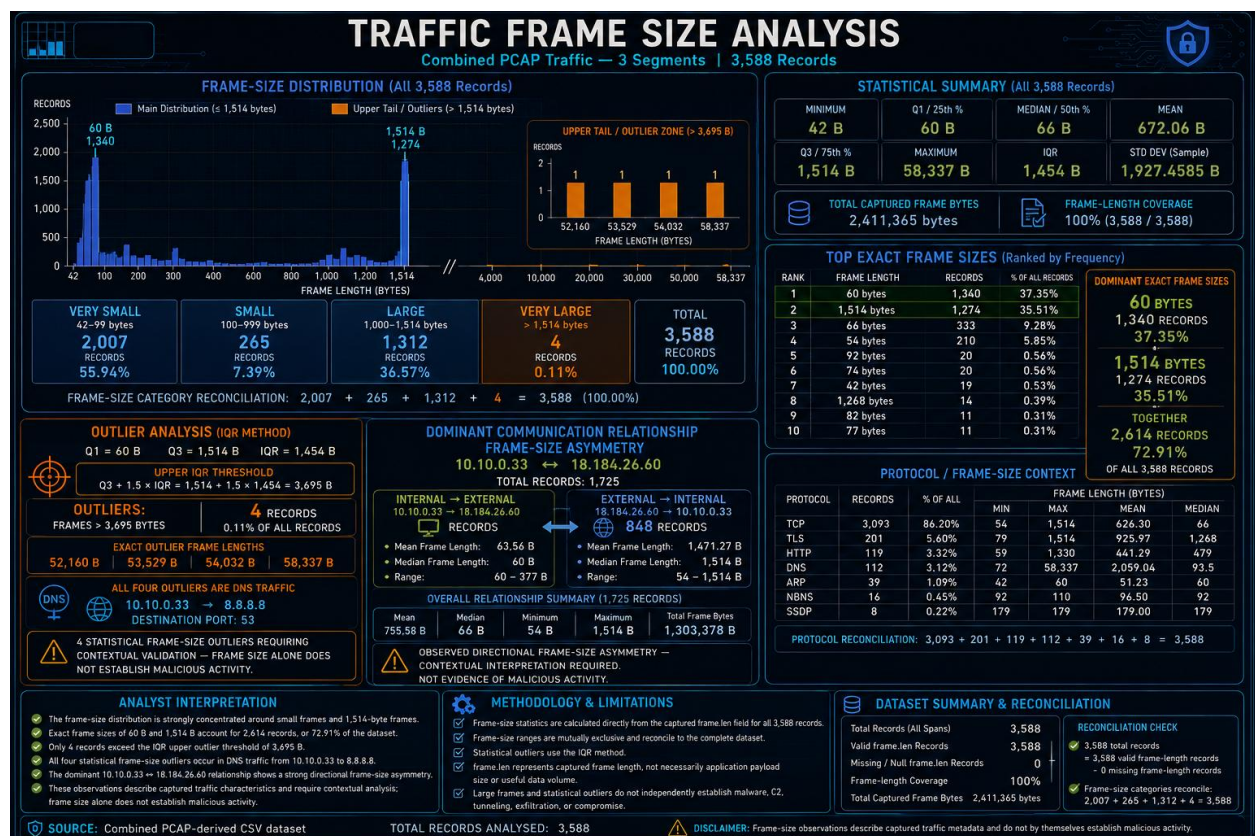
Routine / Observed Traffic

The following traffic was treated separately: ARP/non-IP traffic, accounting for 39 records (1.09%); NBNS local broadcast traffic from 10.10.0.33 to 10.10.0.255 over UDP/137, accounting for 16 records (0.45%); and SSDP multicast traffic from 10.10.0.33 to 239.255.255.250 over UDP/1900, accounting for 8 records (0.22%). Broadcast and multicast traffic were excluded from the public-unicast totals.

Investigation-Relevant Observations

The following patterns were identified as requiring contextual investigation: 1,785 internal-to-public-unicast records; 1,740 public-unicast-to-internal records; 3,525 true internal-to-public-unicast records; 1,725 records involving 10.10.0.33 and 18.184.26.60; 804 TCP/53480 records; 1,221 TCP/80 records; 119 HTTP records; 201 TLS records; 112 DNS records; repeated and bursty communication behaviour; and four statistical frame-size outliers. The analysis did not classify these observations as confirmed malicious activity.

10. Frame-Size Analysis



Visual 7: Traffic Frame Size Analysis

Frame-length statistics were calculated using the captured frame.len values.

Statistical Summary

Statistic	Value
Minimum	42 B

Statistic	Value
Q1 / 25th percentile	60 B
Median / 50th percentile	66 B
Mean	672.0638 B
Q3 / 75th percentile	1,514 B
IQR	1,454 B
IQR Upper Threshold	3,695 B
Maximum	58,337 B
Frames Above Threshold	4
Percentage Above Threshold	0.11%

The IQR upper threshold was calculated as Q3 plus 1.5 times the IQR: $1,514 + (1.5 \times 1,454) = 3,695$ bytes. Four records exceeded this threshold, with exact outlier frame lengths of 52,160 bytes, 53,529 bytes, 54,032 bytes, and 58,337 bytes. All four were DNS traffic from 10.10.0.33 to 8.8.8.8 using UDP/53.

Frame-Size Category Reconciliation

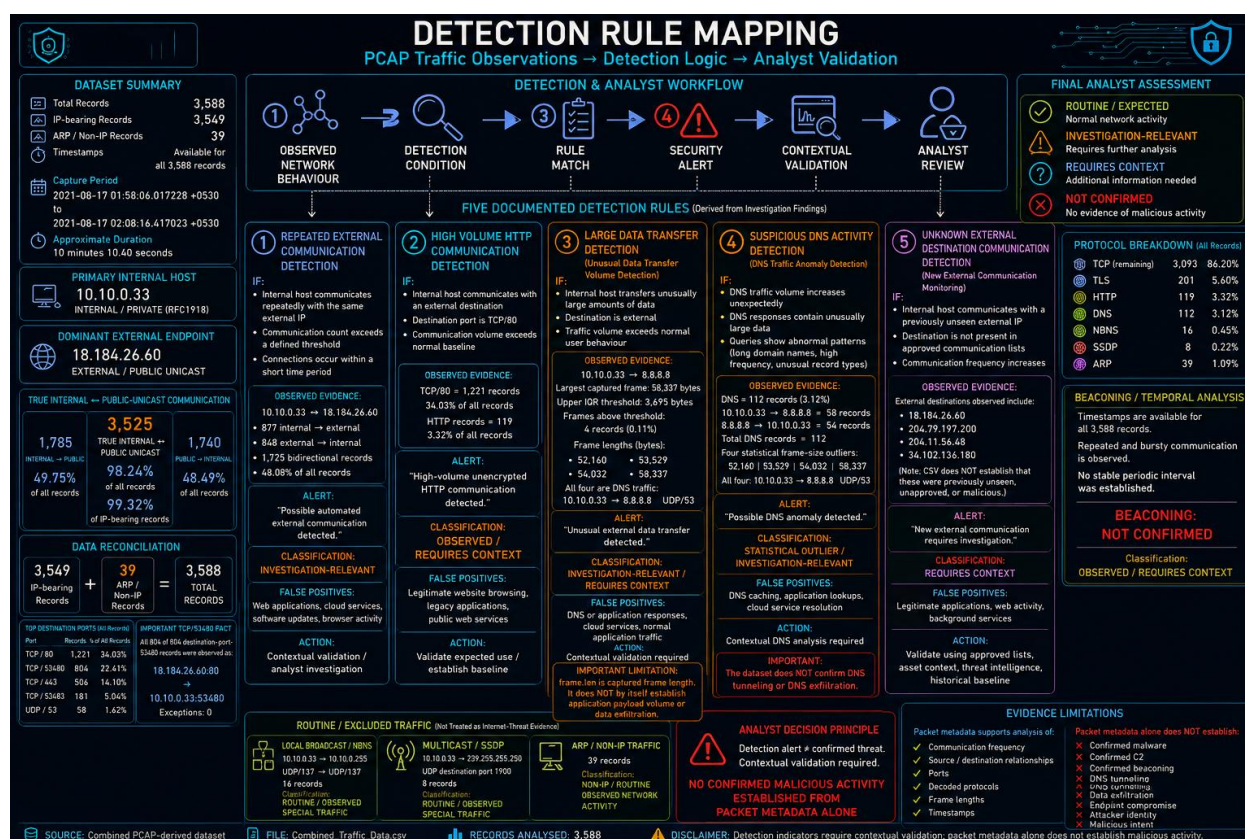
Category	Records	Percentage
42 to 99 bytes	2,007	55.94%
100 to 999 bytes	265	7.39%
1,000 to 1,514 bytes	1,312	36.57%
Above 1,514 bytes	4	0.11%
Total	3,588	100.00%

The reconciliation is straightforward: $2,007 + 265 + 1,312 + 4 = 3,588$.

Interpretation

The four large frames are statistical outliers. However, frame.len represents captured frame length only, and frame length alone does not establish data exfiltration, DNS tunnelling, malware, command-and-control activity, or compromise. The four DNS frame-size outliers therefore require contextual DNS analysis rather than being treated as confirmed malicious traffic.

11. Detection Rule Development



Visual 8: Detection Rule Mapping

Based on the combined traffic analysis, five detection-rule themes were developed.

Rule 1: Repeated External Communication Detection

IF:

An internal host communicates repeatedly with the same external IP
AND communication count exceeds an established threshold
AND connections occur repeatedly within a relatively short period

THEN:

Generate an investigation alert for repeated external communication.

The strongest relationship was 10.10.0.33 to 18.184.26.60, with 1,725 records.

Possible false positives include legitimate web applications, cloud services, software updates, browser activity, and other expected external services.

Analyst action: Validate expected use, ownership, and historical baseline.

Rule 2: High-Volume HTTP / TCP-80 Communication Detection

IF:

An internal host communicates with an external destination
AND destination port is TCP/80

AND communication volume exceeds the expected baseline

THEN:

Generate an investigation alert for high-volume unencrypted HTTP communication.

TCP/80 accounted for 1,221 records (34.03% of all records), and HTTP accounted for 119 records (3.32% of all records).

Possible false positives include legitimate websites, legacy applications, public web services, and normal browser activity.

Analyst action: Validate expected application use and baseline behaviour.

Rule 3: Large Frame-Size Outlier Detection

IF:

An internal host generates unusually large frames
AND frame length exceeds the statistical upper threshold
AND the destination is external or otherwise requires investigation

THEN:

Generate an investigation alert for unusually large frame-size activity.

The IQR upper threshold was 3,695 bytes, and 4 frames exceeded it, with the largest reaching 58,337 bytes. All four statistical outliers were DNS traffic from 10.10.0.33 to 8.8.8.8.

Possible false positives include DNS or application responses, cloud services, normal application traffic, and legitimate large responses.

Analyst action: Perform contextual DNS and traffic analysis.

Rule 4: Suspicious DNS Activity Detection

IF:

DNS traffic volume increases unexpectedly
AND DNS responses contain unusually large data
AND queries display abnormal characteristics such as unusually long names, high frequency, or unusual record types

THEN:

Generate an investigation alert for possible DNS anomaly activity.

DNS traffic accounted for 112 records (3.12% of all records), including 58 records from 10.10.0.33 to 8.8.8.8 and 54 records from 8.8.8.8 to 10.10.0.33. The four large frame-size statistical outliers described earlier were also observed within this DNS traffic.

Possible false positives include DNS caching, application lookups, cloud service resolution, and legitimate DNS responses. The dataset does not confirm DNS tunnelling or DNS exfiltration.

Analyst action: Perform contextual DNS analysis.

Rule 5: Unknown External Destination Communication Detection

IF:

An internal host communicates with a previously unseen external IP
AND the destination is not present in approved communication lists
AND communication frequency increases

THEN:

Generate an investigation alert for new external communication.

The analysis identified external destinations including 18.184.26.60, 204.79.197.200, 204.11.56.48, and 34.102.136.180. The dataset itself does not establish whether these destinations were previously approved, malicious, or unauthorized.

Analyst action: Validate using approved destination lists, asset context, threat intelligence, and historical baseline.

12. Overall Security Investigation Dashboard



Visual 9: Overall Security Investigation Dashboard

The final dashboard consolidates the findings from the entire investigation. The overall posture was investigation required, while also stating no confirmed compromise. The key message is that the traffic contains investigation-worthy indicators, but confirmed malicious activity was not established from available packet metadata alone.

12.1 Key Investigation Findings

The dashboard consolidates the following major findings:

Finding	Detail
Internal to Public Communication	3,525 records (98.24% of all records, 99.32% of IP-bearing records)
Dominant Relationship	10.10.0.33 to 18.184.26.60, 1,725 records
TCP/53480	804 records (22.41%); all associated with 18.184.26.60:80 to 10.10.0.33:53480, 0 exceptions
DNS	112 records (3.12%); activity involving 10.10.0.33 and 8.8.8.8
Frame-Size Outliers	4 records above the 3,695-byte IQR threshold; largest frame 58,337 bytes

13. Investigation Priorities

The final dashboard presents the investigation priorities in the following order.

Priority 1: 10.10.0.33

The primary internal/private host was the dominant communicating internal host and participated in the observed public-unicast communication. Recommended validation includes endpoint and process investigation.

Priority 2: 18.184.26.60

This was the dominant external endpoint. The relationship between 10.10.0.33 and 18.184.26.60 contained 1,725 records, so threat intelligence, ownership, and expected-use validation are appropriate.

Priority 3: DNS Activity Involving 8.8.8.8

The dataset contained 112 DNS records and four large frame-size statistical outliers associated with DNS traffic. Further DNS analysis is required.

Priority 4: TCP/53480 Communication

TCP/53480 contained 804 records, representing 22.41% of the total dataset. The concentration of this communication makes it investigation-relevant.

Priority 5: Segment 3 External Communication

The final dashboard identifies communication involving 204.79.197.200 as an additional investigation item based on the previous Segment 3 analysis.

14. AI-Assisted Visual Analysis

AI tools were used during Milestone 3 to assist with the creation, interpretation, and review of the network traffic visualizations. The purpose of this assistance was to identify important statistical patterns, interpret network communication relationships, highlight potential anomalies, map observations to detection rules, explain possible false positives, assist with comparative analysis, and consolidate findings into a final investigation dashboard.

AI-generated visuals were not treated as independent evidence. The underlying statistical values were required to remain consistent with the previously validated dataset findings.

15. AI Reasoning Verification

The AI-generated visual interpretations were reviewed against the established traffic statistics and previous investigation findings. Important reconciliation checks included the following.

Check	Calculation
Dataset reconciliation	3,549 IP-bearing + 39 ARP/non-IP = 3,588 total records
Public-unicast reconciliation	1,785 Internal -> Public + 1,740 Public -> Internal = 3,525 records
Percentage reconciliation	3,525 / 3,588 = 98.24%; 3,525 / 3,549 = 99.32%
Dominant relationship	877 + 848 = 1,725 (10.10.0.33 and 18.184.26.60)
TCP/53480 verification	804 of 804 records = 18.184.26.60:80 to 10.10.0.33:53480, 0 exceptions
Protocol reconciliation	3,093 + 201 + 119 + 112 + 39 + 16 + 8 = 3,588
Frame-size reconciliation	2,007 + 265 + 1,312 + 4 = 3,588
Outlier verification	4 records above 3,695 bytes, all DNS traffic from 10.10.0.33 to 8.8.8.8

These checks helped verify that the AI-generated visual interpretations remained aligned with the established investigation results.

16. Comparative Analysis: Normal vs Investigation-Relevant Traffic

The investigation distinguished between traffic that was observed as routine/special-purpose and traffic requiring further investigation.

Routine / Observed Traffic

The following were treated as routine or special-purpose network activity: ARP/non-IP traffic (39 records), NBNS local broadcast traffic (16 records), and SSDP multicast traffic (8 records). These categories were separated from public-unicast internet communication.

Investigation-Relevant Traffic

The following observations warranted further analysis: high internal-to-public communication volume; the dominant external endpoint relationship; high-frequency TCP/80 communication; high-

frequency TCP/53480 communication; DNS activity involving 8.8.8.8; four large DNS frame-size outliers; repeated and bursty communication; and external destinations requiring contextual validation.

The important distinction is that investigation-relevant traffic is not the same as confirmed malicious traffic: an observation can trigger an investigation without providing sufficient evidence to classify the traffic as malicious.

17. Evidence Limitations

The analysis was primarily based on packet metadata contained in the PCAP-derived CSV dataset. The available metadata supports analysis of communication frequency, source and destination relationships, IP addresses, ports, decoded protocols, frame lengths, timestamps, and traffic direction.

However, the available metadata alone does not establish confirmed malware, confirmed command-and-control communication, confirmed beaconing, DNS tunnelling, DNS exfiltration, data exfiltration, endpoint compromise, attacker identity, malicious intent, or payload contents. Investigation findings were therefore intentionally classified as observed, investigation-relevant, or requiring context, rather than automatically being labelled malicious.

18. Recommended Next Actions

Based on the final dashboard and detection analysis, the following actions were identified.

1. Threat Intelligence Validation

Validate relevant external IP addresses, particularly 18.184.26.60, 204.79.197.200, 88.221.0.167, 204.11.56.48, and 34.102.136.180, using ownership, reputation, and threat-intelligence information.

2. DNS Investigation

Review the DNS activity involving 10.10.0.33 and 8.8.8.8, giving particular attention to the four large DNS frame-size outliers. The objective is to determine whether the observed DNS traffic represents normal DNS/application behaviour or requires additional investigation.

3. TCP Stream / Payload Inspection

The TCP/53480 relationship (18.184.26.60:80 to 10.10.0.33:53480) should be reconstructed where possible, to determine what data was actually exchanged rather than relying solely on port and packet-count information.

4. Endpoint / Process Investigation

The internal host 10.10.0.33 should be reviewed using endpoint telemetry, process information, and application context where available. This can help determine which local application generated the observed external communication.

5. Baseline and Security-Log Review

Traffic should be compared against normal host behaviour and available firewall logs, DNS logs, proxy logs, EDR/security alerts, endpoint telemetry, and historical network baseline. This would provide the context required to determine whether the observed traffic was expected.

19. Final Analyst Assessment

The combined PCAP-derived dataset contained 3,588 records across the three Milestone 2 traffic segments. The analysis identified a strong concentration of communication between the internal/private host 10.10.0.33 and public-unicast destinations: a total of 3,525 records represented true internal-to-public-unicast communication, corresponding to 98.24% of all records and 99.32% of IP-bearing records.

The strongest communication relationship was 10.10.0.33 to 18.184.26.60, with 1,725 records. Additional investigation-relevant observations included 1,221 TCP/80 records, 804 TCP/53480 records, 506 TCP/443 records, 112 DNS records, four frame-size outliers above 3,695 bytes, and repeated and bursty communication.

The four large frame-size outliers were DNS traffic involving 10.10.0.33 and 8.8.8.8. The analysis did not establish a stable periodic interval sufficient to confirm beaconing, and the available packet metadata did not independently establish malware, command-and-control activity, DNS tunnelling, data exfiltration, endpoint compromise, or malicious intent.

The final security posture is therefore investigation required, with no confirmed compromise. The observations should be treated as indicators requiring contextual validation rather than as confirmed malicious activity.

20. Conclusion

Milestone 3 successfully transformed the network traffic observations from the previous PCAP investigation into a structured detection and visualization workflow. The three Milestone 2 traffic segments were first combined using Python into `Combined_Traffic_Data.csv`, containing 3,588 records.

The combined dataset was then used consistently for statistical analysis, network traffic flow analysis, protocol distribution analysis, destination-port analysis, top-talker analysis, external communication analysis, routine versus investigation-relevant comparison, frame-size analysis, detection-rule development, AI-assisted visual interpretation, AI reasoning verification, and the overall security assessment.

Nine finalized visuals were produced to communicate the findings: network traffic flow analysis, protocol distribution analysis, destination port usage analysis, top communicating hosts, external communication analysis, routine versus investigation-relevant traffic, traffic frame size analysis, detection rule mapping, and the overall security investigation dashboard.

The visualizations and statistical analysis demonstrate that the captured traffic contains several investigation-relevant characteristics, particularly concentrated internal-to-external communication, the dominant 10.10.0.33 to 18.184.26.60 relationship, high-frequency TCP/80 and TCP/53480 traffic,

DNS activity, and large DNS frame-size outliers. At the same time, the investigation maintained an important distinction between anomaly/investigation indicators and confirmed malicious activity.

The final conclusion is that the available packet metadata identifies traffic that warrants contextual security investigation, but it does not independently establish confirmed malicious activity or compromise. Further determination would require additional contextual evidence such as payload inspection, endpoint telemetry, DNS analysis, threat intelligence, security logs, and baseline comparison.